

Rapid Sequencing Kit V14 - gDNA (SQK-RAD114)

Overview

Before beginning, we strongly recommend that first-time users read and review the complete online protocol on the Nanopore Community. The online protocol provides the full instructions, additional commentary, and considerations for various steps, as well as guidance on input amounts, DNA handling, and sequencing. **This reference card is designed for experienced users.**

1. Library preparation (~10 minutes)

- 1 Once thawed, keep all the kit components on ice.
- 2 Prepare the DNA in nuclease-free water.
 - Transfer 100-150 ng genomic DNA into a 1.5 ml Eppendorf DNA LoBind tube.
 - Adjust the volume to 10 µl with nuclease-free water.
 - Mix by flicking the tube to avoid unwanted shearing.
 - Spin down briefly in a microfuge.
- 3 In a 0.2 ml thin-walled PCR tube, mix the following:
- 4 Mix gently by flicking the tube, and spin down.
- 5 Incubate the tube at 30°C for 2 minutes and then at 80°C for 2 minutes. Briefly put the tube on ice to cool it down.
- 6 In a fresh 1.5 ml Eppendorf DNA LoBind tube, dilute the Rapid Adapter (RA) as follows and pipette mix:
- 7 Add 1 µl of diluted Rapid Adapter (RA) to the tagmented DNA.
- 8 Mix gently by flicking the tube, and spin down.
- 9 Incubate the reaction for 5 minutes at room temperature.

Reagent	Volume
100-150 ng template DNA	10 µl
Fragmentation Mix (FRA)	1 µl
Total	11 µl

Reagents	Volume
Rapid Adapter (RA)	1.5 µl
Adapter Buffer (ADB)	3.5 µl
Total	5 µl

2. Priming and loading the MinION and GridION Flow Cell (~5 minutes)

- 1 Thaw the Sequencing Buffer (SB), Library Beads (LIB) or Library Solution (LIS, if using), Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.
- 2 To prepare the flow cell priming mix with BSA, combine Flow Cell Flush (FCF) and Flow Cell Tether (FCT), as directed below. Mix by pipetting at room temperature.

In a suitable tube for the number of flow cells, combine the following reagents:

Reagent	Volume per flow cell
Flow Cell Flush (FCF)	1,170 µl
Bovine Serum Albumin (BSA) at 50 mg/ml	5 µl
Flow Cell Tether (FCT)	30 µl
Total volume	1,205 µl

- 3 Open the MinION or GridION device lid and slide the flow cell under the clip. Press down firmly on the priming port cover to ensure correct thermal and electrical contact.
- 4 Slide the flow cell priming port cover clockwise to open the priming port.
- 5 After opening the priming port, check for a small air bubble under the cover. Draw back a small volume to remove any bubbles:

1. Set a P1000 pipette to 200 µl
2. Insert the tip into the priming port
3. Turn the wheel until the dial shows 220-230 µl, to draw back 20-30 µl, or until you can see a small volume of buffer entering the pipette tip

Note: Visually check that there is continuous buffer from the priming

port across the sensor array.

- 6 Load 800 µl of the priming mix into the flow cell via the priming port, avoiding the introduction of air bubbles. Wait for five minutes. During this time, prepare the library for loading by following the steps below.
- 7 Thoroughly mix the contents of the Library Beads (LIB) by pipetting.
- 8 In a new 1.5 ml Eppendorf DNA LoBind tube, prepare the library for loading as follows:

Reagent	Volume per flow cell
Sequencing Buffer (SB)	37.5 µl
Library Beads (LIB) mixed immediately before use, or Library Solution (LIS), if using	25.5 µl
DNA library	12 µl
Total	75 µl

- 9 Complete the flow cell priming:
 1. Gently lift the SpotON sample port cover to make the SpotON sample port accessible.
 2. Load **200 µl** of the priming mix into the flow cell priming port (**not** the SpotON sample port), avoiding the introduction of air bubbles.
- 10 Mix the prepared library gently by pipetting up and down just prior to loading.
- 11 Add 75 µl of the prepared library to the flow cell via the SpotON sample port in a dropwise fashion. Ensure each drop flows into the port before adding the next.
- 12 Gently replace the SpotON sample port cover, making sure the bung enters the SpotON port and close the priming port.

- 13 Place the light shield onto the flow cell, as follows:

1. Carefully place the leading edge of the light shield against the clip.

Note: Do not force the light shield underneath the clip.

2. Gently lower the light shield onto the flow cell. The light shield should sit around the SpotON cover, covering the entire top section of the flow cell.

3. Flow cell reuse and returns

- 1 After your sequencing experiment is complete, if you would like to reuse the flow cell, please follow the Flow Cell Wash Kit protocol and store the washed flow cell at +2°C to +8°C.

The [Flow Cell Wash Kit protocol](#) is available on the Nanopore Community.

- 2 Alternatively, follow the returns procedure to send the flow cell back to Oxford Nanopore. Instructions for returning flow cells can be found [here](#).